PRACT 5 Sales

November 2, 2026

[1]:
```
# Step 1: Import required libraries
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.decomposition import PCA
```

[3]:
```
# Step 2: Load the dataset
df = pd.read_csv(r"E:\LP3_LAB Manual\ML PRAC\PRACT5\sales_data_sample (1) -␣
↪sales_data_sample (1).csv", encoding="latin1")

# Step 3: Display the dataset
print("Dataset:")
print(df.head())
```

```
Dataset:
   ORDERNUMBER  QUANTITYORDERED  PRICEEACH  ORDERLINENUMBER    SALES  \
0        10107               30      95.70                2  2871.00
1        10121               34      81.35                5  2765.90
2        10134               41      94.74                2  3884.34
3        10145               45      83.26                6  3746.70
4        10159               49     100.00               14  5205.27

          ORDERDATE   STATUS  QTR_ID  MONTH_ID  YEAR_ID  ...  \
0    2/24/2003 0:00  Shipped       1         2     2003  ...
1    5/7/2003 00:00  Shipped       2         5     2003  ...
2    7/1/2003 00:00  Shipped       3         7     2003  ...
3    8/25/2003 0:00  Shipped       3         8     2003  ...
4  10/10/2003 00:00  Shipped       4        10     2003  ...

                   ADDRESSLINE1  ADDRESSLINE2           CITY STATE  \
0        897 Long Airport Avenue           NaN            NYC    NY
1             59 rue de l'Abbaye           NaN          Reims   NaN
2  27 rue du Colonel Pierre Avia           NaN          Paris   NaN
3             78934 Hillside Dr.           NaN       Pasadena    CA
4                7734 Strong St.           NaN  San Francisco    CA
```

```
  POSTALCODE COUNTRY TERRITORY CONTACTLASTNAME CONTACTFIRSTNAME DEALSIZE
0      10022     USA       NaN              Yu             Kwai    Small
1      51100  France      EMEA         Henriot             Paul    Small
2      75508  France      EMEA        Da Cunha           Daniel   Medium
3      90003     USA       NaN           Young            Julie   Medium
4        NaN     USA       NaN           Brown            Julie   Medium

[5 rows x 25 columns]
```

[5]:
```
# Step 4: Select numerical features
features = [
    "QUANTITYORDERED",
    "PRICEEACH",
    "SALES"
]
X = df[features]

# Step 5: Handle missing values
X = X.dropna()
```

[6]:
```
# Step 6: Standardize the data
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

[7]:
```
# Step 7: Calculate WCSS for different cluster values
wcss = []

for i in range(1, 11):
    kmeans = KMeans(
        n_clusters=i,
        random_state=42,
        n_init=10
    )

    kmeans.fit(X_scaled)

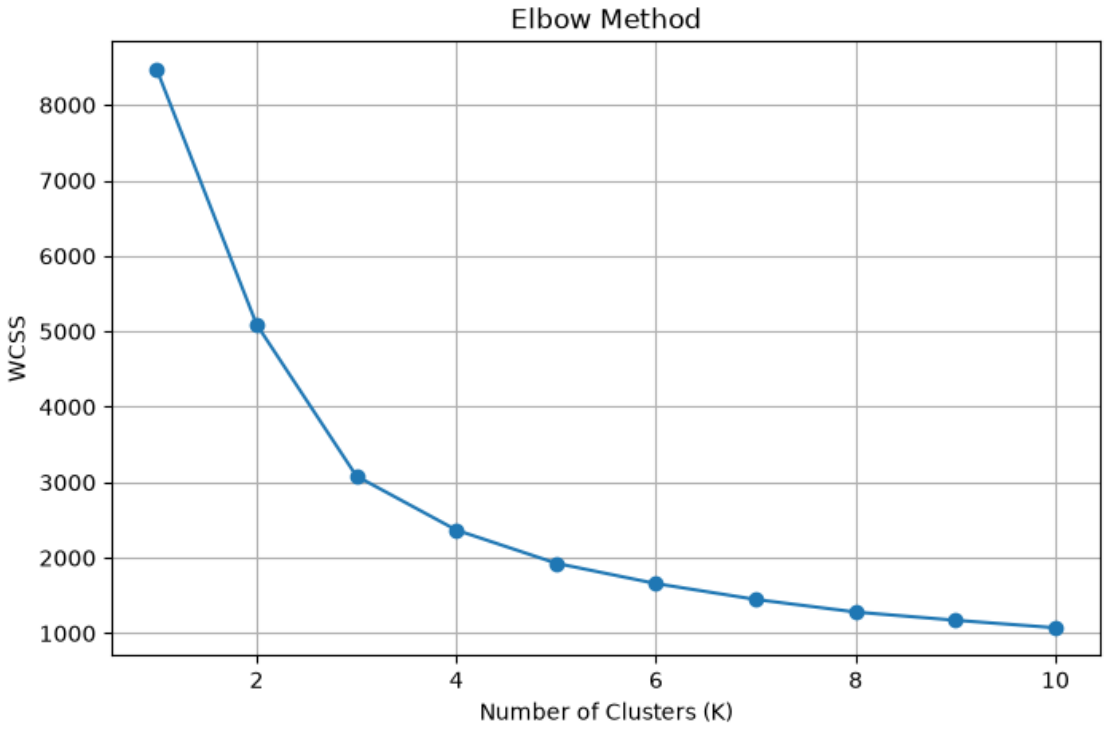
    wcss.append(kmeans.inertia_)
```

[15]:
```
# Step 8: Plot the Elbow Method graph
plt.figure(figsize=(8, 5))

plt.plot(
    range(1, 11),
    wcss,
    marker="o"
)

plt.title("Elbow Method")
```

```
plt.xlabel("Number of Clusters (K)")
plt.ylabel("WCSS")
plt.grid(True)
plt.show()
```



[16]:
```
# Step 9: Choose the optimal number of clusters
# Observe the elbow point in the graph
optimal_k = 3

# Step 10: Apply K-Means clustering
kmeans = KMeans(
    n_clusters=optimal_k,
    random_state=42,
    n_init=10
)

clusters = kmeans.fit_predict(X_scaled)
```

[17]:
```
# Step 11: Add cluster labels to the dataset
df = df.loc[X.index].copy()
df["Cluster"] = clusters
```

[18]:
```
# Step 12: Display clustered data
print("\nClustered Data:")
print(df[features + ["Cluster"]].head(10))
```

```
Clustered Data:
   QUANTITYORDERED  PRICEEACH    SALES  Cluster
0               30      95.70  2871.00        2
1               34      81.35  2765.90        2
2               41      94.74  3884.34        1
3               45      83.26  3746.70        1
4               49     100.00  5205.27        1
5               36      96.66  3479.76        2
6               29      86.13  2497.77        2
7               48     100.00  5512.32        1
8               22      98.57  2168.54        2
9               41     100.00  4708.44        1
```

[19]:
```
# Step 13: Display number of customers in each cluster
print("\nCustomers in Each Cluster:")
print(df["Cluster"].value_counts())
```

```
Customers in Each Cluster:
Cluster
2    1035
1     907
0     881
Name: count, dtype: int64
```

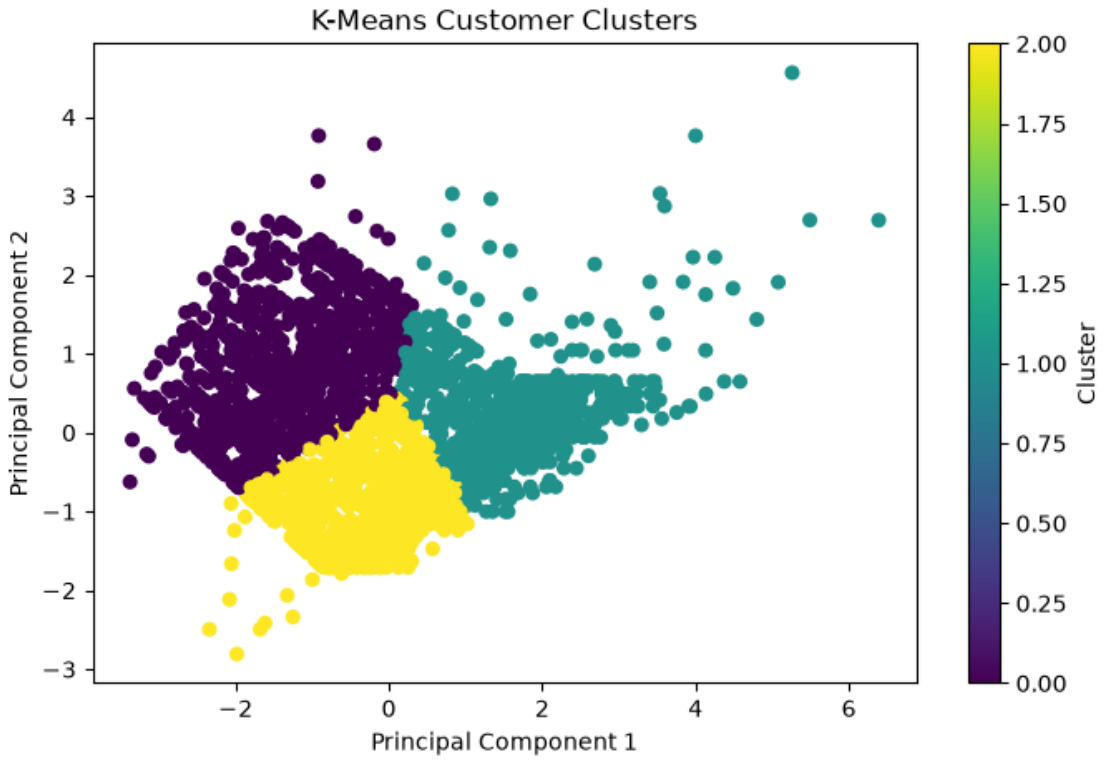
[20]:
```
# Step 14: Visualize clusters using PCA
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)

plt.figure(figsize=(8, 5))

plt.scatter(
    X_pca[:, 0],
    X_pca[:, 1],
    c=clusters,
    cmap="viridis",
    s=30
)

plt.title("K-Means Customer Clusters")
plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.colorbar(label="Cluster")
```

```
plt.show()
```


K-Means Customer Clusters
Principal Component 2
Principal Component 1
Cluster
4
3
2
1
0
−1
−2
−3
−2
0
2
4
6
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00

[ ]:

[ ]: